USAT COLLEGE SAGAY CITY, INC.

BACHELOR OF SCIENCE IN INFORMATION SYSTEMS — DATA MINING & EXPLORATION
Part 2: Descriptive Statistics, Data Visualization, and Activity Report

STUDENT ID	STUDY HOURS	QUIZ SCORE	LEARNING PREFERENCE
S01	2	70	Online
S02	3	75	Hybrid
S03	4	75	Online
S04	5	80	Face-to-Face
S05	6	85	Hybrid
S06	3	75	Online
S07	7	90	Face-to-Face
S08	4	80	Online
S09	8	95	Hybrid
S10	2	65	Face-to-Face

INDIVIDUAL ACTIVITY: GUIDE QUESTIONS & ANSWERS

Q1. What is the typical quiz score based on the mean and median? Are they similar?

Answer:

• Mean Quiz Score = 80.00
• Median Quiz Score = 77.50

Q2. Which quiz score occurs most frequently?

Answer:

• Mode Quiz Score = 75

The score of 75 occurs 3 times in the dataset (S02, S03, and S06), making it the most frequent

Q3. Which learning preference is most common? How many students selected it?

Answer:

• Most Common Preference = Online

A total of 4 students (S01, S03, S06, S08) selected 'Online'. Hybrid was selected by 3 students,

Q4. What pattern, if any, appears between study hours and quiz scores?

Answer:

• Strong Positive Linear Pattern / Association:

As study hours increase, quiz scores consistently increase. Students with lower study hours (e

Q5. Does the scatter plot prove that studying longer causes higher scores? Explain.

Answer:

• No, it does NOT prove causation.

As stated in Lesson 3, 'Association does not establish causation.' The scatter plot demonstrat

Q6. What is one limitation of drawing conclusions from only 10 fictional records?

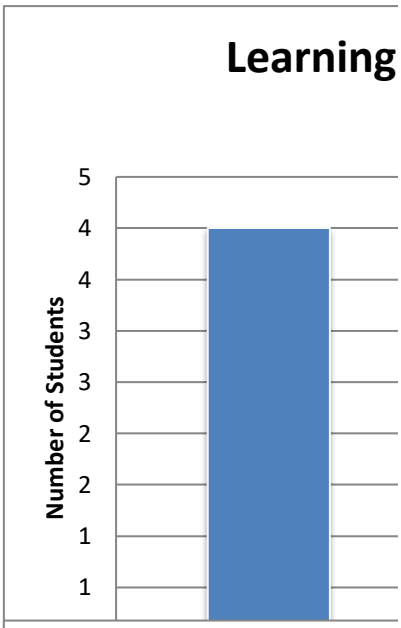
Answer:

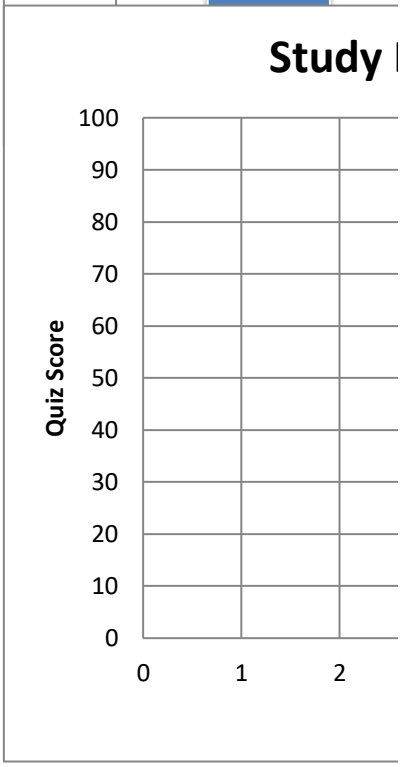
• Limitation (Small Sample Size & Lack of Generalizability):

A sample size of N=10 is extremely small and highly sensitive to individual outliers. Findings f

DESCRIPTIVE STATISTICS (QUIZ SCORE)	
Mean (Average)	**79.00**
Median (Middle Value)	**77.5**
Mode (Most Frequent)	**#NAME?**
Minimum	**65**
Maximum	**95**
Range (Max - Min)	**30**

FREQUENCY DISTRIBUTION	
Learning Preference	**Frequency**
Online	4
Hybrid	3
Face-to-Face	3
Total	**10**





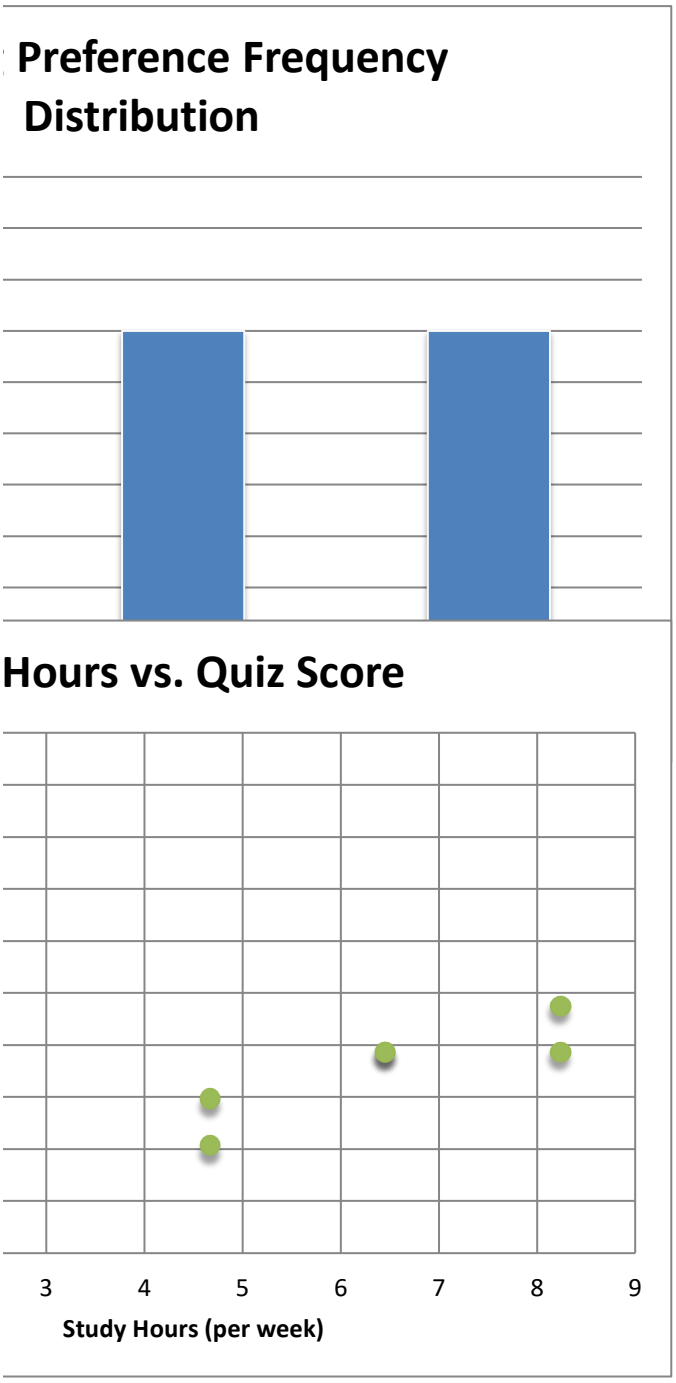
t score.

, and Face-to-Face was selected by 3 students.

e.g., 2 hours) scored lower (65-70), while students studying

es a strong positive correlation, but correlation alone does not

rom 10 fictional records lack statistical power and


Preference Frequency Distribution
Hours vs. Quiz Score
3
4
5
6
7
8
9
Study Hours (per week)